

Behavioral Research Workflow Workbook

A hands-on companion to the workshop deck – setup, prompt templates, and troubleshooting

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How This Workbook Is Organized

Sections 1-6 walk through one concrete, already-completed case study – a temporal reframing savings A/B test (“\$5 a day” vs. “\$150 a month” framing) – built and run end-to-end with this project’s own tools: **VS Code + Claude Code** for the agentic workflow, **Python** (pandas, numpy, scipy, statsmodels) for sampling and analysis, **git/GitHub** for version control, and **pandoc** for generating this workbook, the workshop deck, and the study writeup. Those sections assume the four Skills this case study used (Stages 3, 4, 5, 7) already exist, because they do, in this repository – you’re reading how they were used, not how to build them.

Section 7 (Appendix) zooms out from that one case study: it’s a general guide to setting up this kind of **9-stage** workflow for a *different* research topic, starting from a bare repository where none of those skill files exist yet. Read Sections 1-6 first to see the pattern in action end-to-end; read the Appendix when you’re ready to build your own version of it. Section 7.6 also covers a no-cost Google alternative (Antigravity) for students without access to Claude Code.

Quick Start: the whole flow in one page

If you only read one page before opening a terminal, read this one. Everything here is expanded later.

1. **Set up once:** VS Code + Claude Code, Python venv, git/GitHub, pandoc (Section 1).
2. **Scaffold a project:** the folder layout in Section 2 – one projects/<slug>/ per study.
3. **Stage 3 (design):** give Claude your control stimulus, treatment stimulus, population description, and outcome variable. Get back a structured design package.
4. **Stage 4 (synthetic test):** pick a total N and batch size (10 is a safe default), run it, get back `subjects_data.csv`.
5. **Stage 5 (analysis):** balance check -> regression -> chart. Read the balance report before trusting the regression.
6. **Stage 7 (writeup):** assemble Method/Balance/Results into a Word doc, optionally benchmarked against a published study.

Worked example at a glance (the workshop’s own case study, so you know what a completed run looks like):

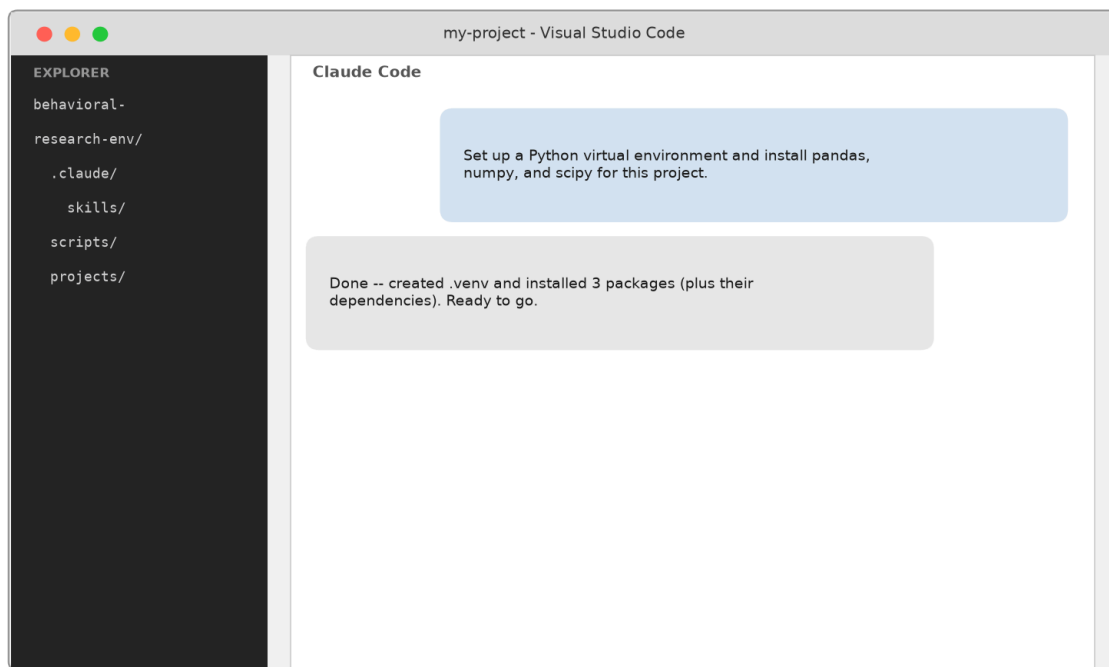
Step	What happened
Design	“\$5 a day” (treatment) vs. “\$150 a month” (control) savings framing; primary outcome = sign-up decision
Pilot	N=20, blinded; 80% vs. 40% sign-up, directionally large but only marginal ($p \sim .08$)
Scale-up	N=400 after unblinding; 64.5% vs. 45.5% sign-up, $B=0.82$, $p < .001$
Writeup	Compared against a real published study: savings-decision effect replicated, affordability effect replicated at about half the size, understandability effect did not replicate

That last row is the point worth remembering: a synthetic run can look convincing and still miss part of the real psychology. Treat it as a fast way to iterate on a design, not a replacement for real data.

1. Prerequisites & Setup

This section gets you from a bare machine to a working environment: VS Code, Claude Code, Python, and git/GitHub.

In plain English: install VS Code and the Claude Code extension, then tell Claude Code what you need – it runs the actual setup commands for you.



Illustrative mockup -- not an actual screenshot

Everything from here through the end of this section is the “**under the hood**” detail: the exact commands Claude Code runs on your behalf. You don’t need to type any of it yourself – it’s here so you understand what’s happening, and so you have it if something needs troubleshooting.

1.1 Install VS Code and Claude Code (*under the hood*)

1. Install **VS Code** from its official source.
2. Install the **Claude Code** extension (or CLI) and sign in.
3. Open a folder in VS Code – this becomes your project’s working directory.

1.2 Install Python (*under the hood*)

Check whether Python is already installed (same command on both platforms):

```
python --version
```

If it’s missing, install it:

Windows (Windows sometimes shows a Microsoft Store stub instead of a real interpreter):

```
winget install Python.Python.3.12
```

Gotcha (Windows): after installing anything with winget, PATH changes do not appear in your *current* terminal session automatically. Refresh it manually, or open a new terminal window:

```
$machinePath = [System.Environment]::GetEnvironmentVariable("Path","Machine")
$userPath = [System.Environment]::GetEnvironmentVariable("Path","User")
$env:Path = $machinePath + ";" + $userPath
```

macOS (via Homebrew – see `brew.sh` if you don’t have it yet):

```
brew install python@3.12
```

Gotcha (macOS): this gotcha is about Homebrew itself, not Python – if commands you just installed aren't found right after *installing Homebrew*, its own bin directory (/opt/homebrew/bin on Apple Silicon, /usr/local/bin on Intel) isn't on your PATH yet. The Homebrew installer prints an eval "\$(brew shellenv)" line to add to ~/.zshrc; add it, then open a new terminal or run source ~/.zshrc.

1.3 Create a project-local virtual environment (*under the hood*)

Windows (PowerShell):

```
python -m venv .venv
.\.venv\Scripts\python.exe -m pip install --upgrade pip
.\.venv\Scripts\python.exe -m pip install -r scripts/requirements.txt
```

macOS (Terminal):

```
python3 -m venv .venv
.venv/bin/python -m pip install --upgrade pip
.venv/bin/python -m pip install -r scripts/requirements.txt
```

(python3 because a bare python isn't guaranteed to exist on macOS. Alternatively, source .venv/bin/activate once, then just call python/pip directly for the rest of the session.)

A minimal requirements.txt for this kind of workflow:

```
pandas
numpy
scipy
statsmodels
matplotlib
openpyxl
python-dotenv
tabulate
```

(Add anthropic only if you plan to call the Claude API directly from a script, rather than using Claude Code subagents.)

1.4 Install git and set up GitHub (*under the hood*)

Windows (PowerShell):

```
winget install --id Git.Git
winget install --id GitHub.cli --source winget --scope user
```

Gotcha (Windows): some winget installers try to trigger an admin-elevation (UAC) prompt, which silently fails in a non-interactive terminal. Adding --scope user installs to your user profile instead and avoids the prompt entirely.

macOS (Terminal):

```
brew install git gh
```

Gotcha (macOS): macOS ships no git binary until either the Xcode Command Line Tools or Homebrew's own git are installed. If running git for the first time pops up a "Install Command Line Tools" prompt, accept it and wait for it to finish – or install git via Homebrew above to skip that prompt entirely.

Authenticate GitHub (this step is interactive – run it yourself in a real terminal, not through an automation tool):

```
gh auth login
```

Then create and push your repo:

```
git init
git add .
git commit -m "Initial commit"
gh repo create <repo-name> --private --source=. --remote=origin --push
```

1.5 Install pandoc (for generating docx/pptx/pdf output) (*under the hood*) Windows (PowerShell):

```
winget install --id JohnMacFarlane.Pandoc --scope user
```

PDF conversion additionally needs a rendering engine. A lightweight option:

```
winget install --id Typst.Typst --scope user
```

macOS (Terminal):

```
brew install pandoc typst
```

Then convert with (same command on both platforms):

```
pandoc mydoc.md -o mydoc.pdf --pdf-engine=typst
```

Gotcha: if your markdown embeds images by relative path, pandoc resolves paths relative to your *current directory*, not the markdown file's location. Fix with `--resource-path`, or by running pandoc from the file's own directory. The flag's syntax differs by platform: Windows uses `--resource-path=".;path\to\assets"`; macOS/Linux use `--resource-path=".:path/to/assets"`.

2. Project Scaffolding

A recommended directory layout for this kind of workflow:

```
my-research-env/
├── .claude/
├── skills/
│   ├── behavioral-design/SKILL.md
│   ├── synthetic-ab-test/SKILL.md
│   ├── data-analysis/SKILL.md
│   └── research-writeup/SKILL.md
└── scripts/
```

```

requirements.txt
synthetic_sample/
  sample_population.py
  persona_prompt.py
  build_batches.py
  merge_responses.py
analysis/
  balance_check.py
  regression.py
  charts.py
projects/
  <study-slug>/
    03_design/
      research_design.md
      variables_spec.md
      population_spec.md
      stimuli/
        control.md
        treatment.md
    04_synthetic_test/
      population_config.json
      test_config.json
      subjects_frame.csv
      subjects_data.csv
    05_analysis/
      balance_report.md
      regression_results_*.md
      charts/*.png
    07_writeup/
      writeup.md
      writeup.docx
.gitignore

```

A `.gitignore` that keeps machine-specific and secret-bearing files out of version control:

```

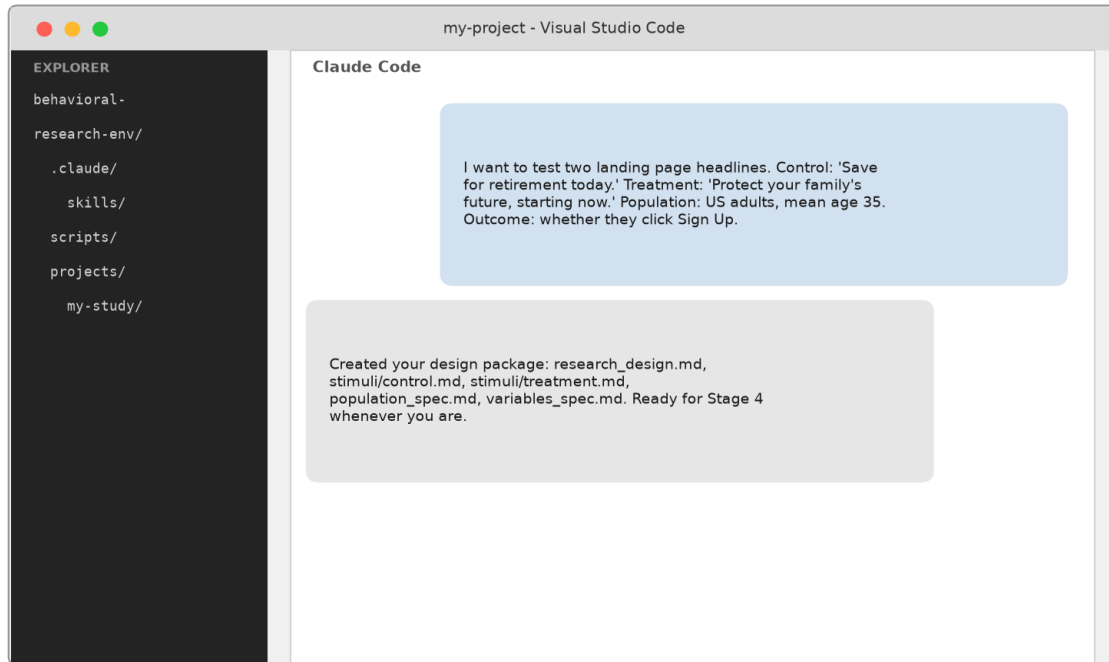
.venv/
__pycache__/
*.pyc
.env

```

3. Stage-by-Stage Guide

Each stage below is a Claude Code **Skill** – a markdown instruction file the AI agent follows. You invoke a stage by describing what you want in plain language; Claude Code figures out which skill applies (or you can name it directly, e.g. “run behavioral-design”). This section assumes those skill files already exist, because they do in this project – if you’re setting this up from a bare repository for a different research topic, see Section 7 for how to create them.

3.1 Stage 3 – Behavioral Design (intake)



Illustrative mockup -- not an actual screenshot

What it does: takes your control stimulus, treatment stimulus, population description, and outcome variable(s), and turns them into a structured design package Stage 4 can consume.

Copy-paste prompt template:

I want to set up a behavioral test.

Control condition stimulus: [PASTE CONTROL STIMULUS TEXT]

Treatment condition stimulus: [PASTE TREATMENT STIMULUS TEXT]

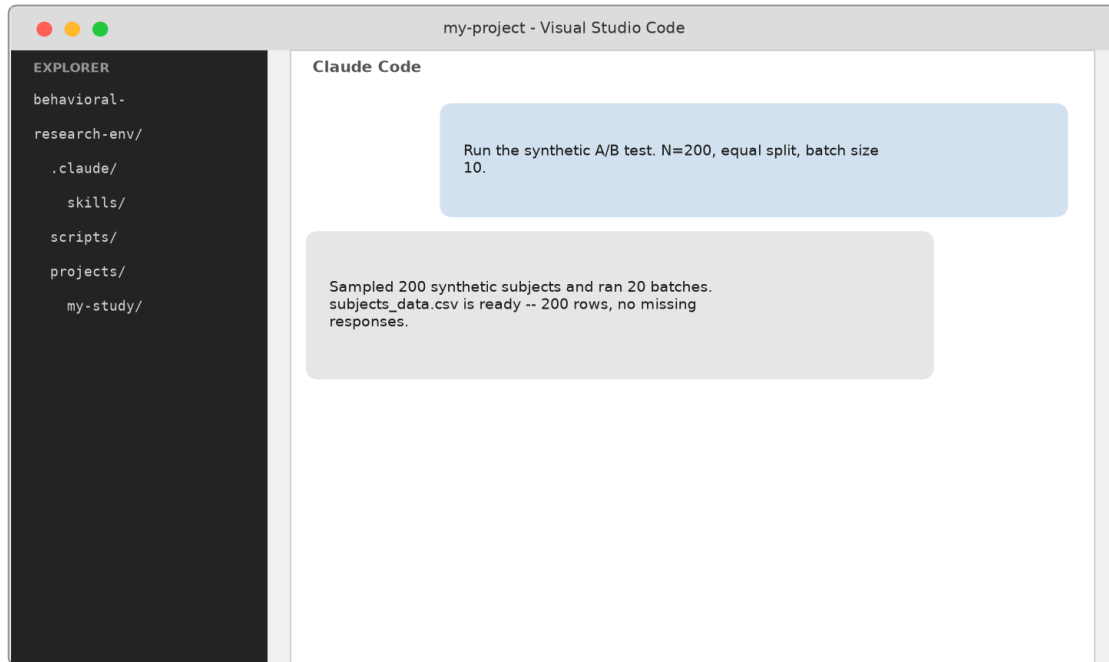
Population to sample: [DESCRIBE DEMOGRAPHICS AND INDIVIDUAL DIFFERENCES – e.g. age mean/SD, gender split, income, education, financial literacy, numeracy – with means/SDs if you have them, or a rough text description if not]

Outcome variable(s): [NAME AND TYPE – e.g. “a binary choice between X and Y” or “a 1-7 Likert rating of Z”]

Please set this up as a new project called [PROJECT-SLUG].

What you get back: research_design.md, stimuli/control.md and stimuli/treatment.md, population_spec.md, variables_spec.md – review these before moving on, especially variables_spec.md’s note on which category should be coded 1 for any binary outcome.

3.2 Stage 4 – Synthetic A/B Test



Illustrative mockup -- not an actual screenshot

What it does: samples a synthetic subject population, randomly assigns them to conditions, and collects each subject's response by simulating them with Claude (subagents, batched) or the direct API.

Copy-paste prompt template:

Let's run the synthetic A/B test for [PROJECT-SLUG].

Total N: [N]. Allocation: [equal / specify counts per condition].

Use subagent mode with batch size [10 is a good default – smaller batches are more reliable per-subject, larger batches are more token-efficient].

What you get back: subjects_frame.csv (sampled profiles + condition), subjects_data.csv (profiles + condition + responses) – spot-check a few rows before moving to analysis.

Under the hood: config file reference and a worked example

Everything below is what Claude Code writes and reads behind the scenes – useful if you want to inspect or hand-edit a config, not required to run the stage.

population_config.json field reference:

Field	Meaning
conditions	list of condition names
n_total	total sample size
allocation	"equal" or a dict of exact per-condition counts
variables	list of {name, kind: continuous\ categorical, role, mean/sd/min/max/round} or {categories, probabilities}
correlations	optional list of {var1, var2, rho} pairs

test_config.json field reference:

Field	Meaning
mode	"subagent" (default, no API key) or "api" (opt-in, needs ANTHROPIC_API_KEY)
conditions	dict of condition name -> stimulus text
outcome_variable	{name, type, options/scale_min/scale_max, description, reasoning_prompt, secondary: [...]}
batch_size	subjects simulated per subagent call (default 10)
agent_type	which subagent type to use (default general-purpose)

Concrete example: a filled-in population_config.json

This is an abbreviated version of the config used for the workshop's own temporal reframing case study (N reduced for space):

```
{
  "conditions": ["condition_1", "condition_2"],
  "n_total": 400,
  "allocation": "equal",
  "variables": [
    {"name": "age", "kind": "continuous", "role": "demographic",
     "mean": 34.29, "sd": 7.24, "min": 18, "max": 85},
    {"name": "gender", "kind": "categorical", "role": "demographic",
     "categories": ["female", "male"], "probabilities": [0.477, 0.523]},
    {"name": "income_bracket", "kind": "continuous", "role": "demographic",
     "mean": 5.29, "sd": 3.2, "min": 1, "max": 16, "round": true},
    {"name": "financial_literacy", "kind": "continuous", "role":
"individual_difference",
     "mean": 2.36, "sd": 0.91, "min": 0, "max": 3, "round": true}
  ]
}
```

```

    ],
    "correlations": []
}

```

And the matching test_config.json:

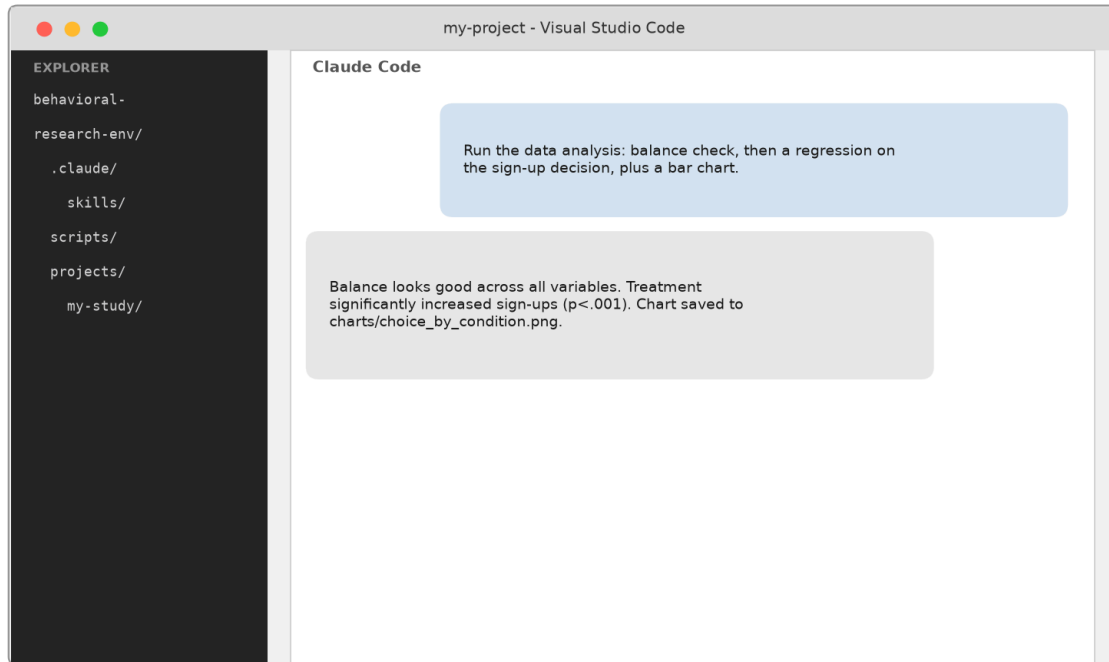
```

{
  "mode": "subagent",
  "conditions": {
    "condition_1": "...get started with $5 a day today...",
    "condition_2": "...get started with $150 a month today..."
  },
  "outcome_variable": {
    "name": "choice",
    "type": "binary",
    "options": ["Not now", "Yes, start saving"],
    "description": "Whether the subject chooses to start saving.",
    "reasoning_prompt": "List three things you were thinking about, in your own words."
  },
  "batch_size": 10,
  "agent_type": "general-purpose"
}

```

Notice round: true on income_bracket and financial_literacy – both are ordinal/count scales in the real instrument (a 16-point income bracket, a 0-3 correct-answer count), so fractional sampled values need rounding to stay meaningful.

3.3 Stage 5 – Data Analysis



Illustrative mockup -- not an actual screenshot

What it does: checks randomization balance, runs regressions on your outcome variable(s), and produces bar charts.

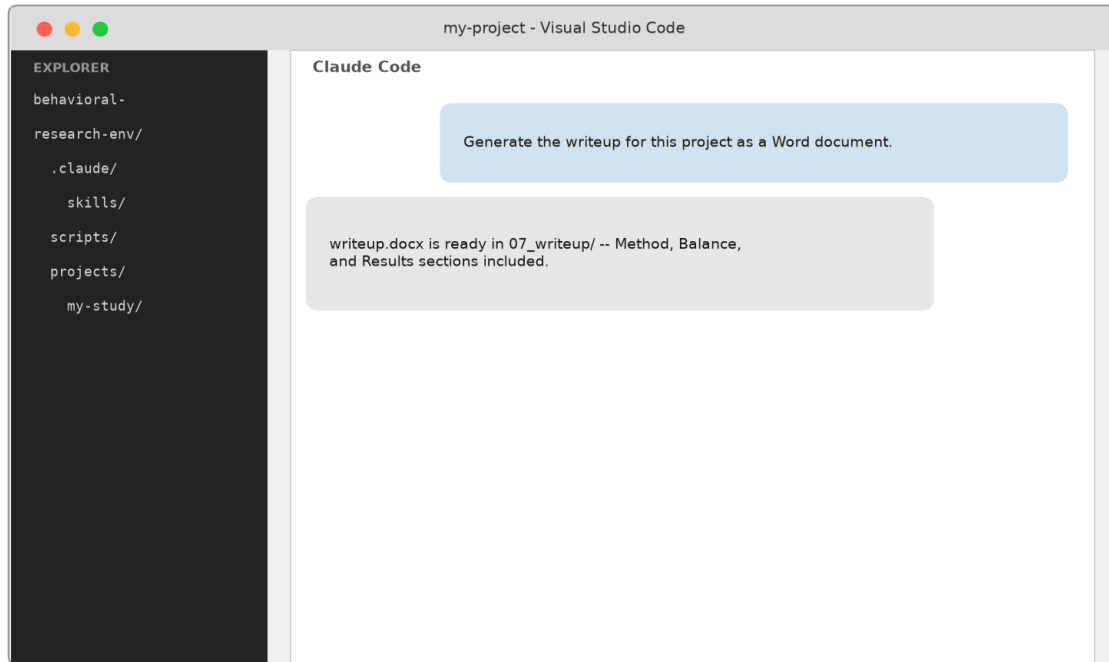
Copy-paste prompt template:

Run the data analysis for [PROJECT-SLUG]:

- Balance check across all sampled demographic/individual-difference variables
- Regression on [OUTCOME VARIABLE NAME], type [binary/continuous], with controls: [none, or list variables]
- If binary, code [WHICH CATEGORY] as 1
- Bar chart of the outcome by condition

What you get back: balance_report.md, regression_results_<outcome>.md, charts/<outcome>_by_condition.png – read the balance report first; if anything is flagged, consider re-running the regression with that variable as a control.

3.4 Stage 7 – Research Writeup



Illustrative mockup -- not an actual screenshot

What it does: assembles Method, Balance, and Results sections into a Word-ready document, optionally compared against a reference study.

Copy-paste prompt template:

Generate a Stage 7 writeup for [PROJECT-SLUG] (results-and-method scope, no mediation/moderation).

[Optional] Compare against this reference study: [CITATION]. Reference statistics: [OUTCOME NAME: effect size, p-value, with/without controls – repeat per outcome you have a reference value for].

What you get back: writeup.md and writeup.docx. If you included a reference comparison, read the classification (replicates / partially replicates / does not replicate) critically – don't let a partial or non-replication get smoothed over in the prose.

4. Critical-Steps Checklist

- **Batch size is a real tradeoff.** Smaller batches (e.g. 2 subjects/call) cost roughly 4x more tokens per subject than larger batches (e.g. 10/call), because fixed prompt overhead (persona instructions, stimuli, schema) gets amortized across more subjects. But very large batches risk subjects' answers drifting toward each other. Start at 10 unless you have a reason not to.

- **Decide your blinding approach before Stage 4.** If you want to avoid analyst bias, ask for neutral condition labels (Condition A/B) and don't reveal which is treatment until after Stage 5 is complete.
- **When balance check flags an imbalance,** don't panic and don't ignore it either – re-run the regression with that variable as a control and confirm the main result holds. With many variables tested, some false positives are expected by chance (roughly a 1-in-4 chance of at least one $p < .05$ false positive across 6 independent tests at $\alpha = .05$).
- **Binary outcome coding is not automatic-safe.** Don't rely on an alphabetical default for which category is coded 1 – state it explicitly, since it determines the sign of every downstream effect.
- **Estimate token cost before scaling up.** At batch size 10, expect roughly 2,500-2,600 tokens per synthetic subject in subagent mode. For N subjects, that's very roughly $N \times 2,600$ tokens total, split across $N / \text{batch_size}$ subagent calls.
- **A pilot run (small N) is worth doing before a full-scale run.** It catches configuration and prompt problems cheaply, and gives you a directional read even if it's not statistically definitive on its own.

4.1 Sizing a run: rough token/effort reference

These are rough, empirically-observed figures from subagent mode at batch size 10 – use them to sanity-check a run's cost/time before launching it, not as exact guarantees.

N (total subjects)	Batches (batch size 10)	Approx. tokens	Notes
20	2	~50,000	good for a quick pilot
100	10	~260,000	moderate live run, still manageable in one session
400	40	~1,040,000	a full-scale run; launch batches in parallel groups (e.g. 6-8 per message) to keep wall-clock time reasonable

For comparison, the same N in **API mode** (opt-in, direct Anthropic API calls, not batched) is roughly 600-800 tokens/subject – cheaper in raw tokens and billed separately in dollars rather than counting against your Claude Code usage, but requires its own API key and billing setup, and doesn't benefit from batching efficiency.

5. Troubleshooting Appendix

Symptom	Cause	Fix
winget install reports success then the tool still isn't found (Windows)	PATH updated on disk but not in your current shell process	Refresh \$env:Path from Machine+User scope, or open a new terminal
winget install fails/cancels with an installer exit code around 1602 (Windows)	Installer tried to trigger a UAC elevation prompt that can't be answered non-interactively	Retry with --scope user
A brew-installed tool isn't found right after installing Homebrew itself (macOS)	Homebrew's own bin directory (/opt/homebrew/bin Apple Silicon, /usr/local/bin Intel) isn't on PATH yet	Add the eval "\$(brew shellenv)" line Homebrew's installer prints to ~/.zshrc, then open a new terminal or source ~/.zshrc
Running git for the first time pops up an "Install Command Line Tools" dialog (macOS)	macOS ships no git binary until Xcode Command Line Tools or Homebrew's git are installed	Accept the prompt and wait for it to finish, or brew install git to skip it
pandoc converts markdown but images are missing from the output	pandoc resolves relative image paths from the working directory, not the markdown file's folder	Add --resource-path (;-separated backslash paths on Windows, :-separated forward-slash paths on macOS/Linux), or cd into the markdown file's folder first
A regression script errors on a column that "should" be numeric	Newer pandas versions (e.g. pandas 1.5.0) store text as a str dtype instead of the classic object dtype, breaking dtype == object checks	Use pandas.api.types.is_numeric_dtype(...) instead of comparing dtype directly
A statistical test (e.g. Bartlett's) throws an obscure internal array error	Some scipy versions mishandle integer-dtype input in certain test implementations	Cast the column to float before passing it to the test
A subagent's JSON response has extra text before/after the array	Subagents don't have a forced-output guarantee the way a direct API tool-call does	Instruct explicitly to return <i>only</i> the JSON array with no surrounding text; strip/retry if it still wraps the output in a code fence
git commit succeeds but shows a warning about name/email being auto-detected	git had no configured identity and guessed one from your username/hostname	Set it explicitly: git config --global user.name "..."; and git config --global user.email "..."; amend the commit with \$-reset-shouldn't do that to explicitly catch the push responsibility and failing
An binary outcome can't recognize insufficient human-made non-invariant sign	The default code in false-back-alphabetical order where no positive category is specified	Always pass the categorical that should be used to explicitly catch the push responsibility and failing

6. Glossary & Command Cheat Sheet

Synthetic subject – an LLM-simulated research participant, given a sampled demographic/individual-difference profile and asked to respond in character.

Batch – a group of synthetic subjects simulated within a single subagent call, to amortize fixed prompt overhead.

Condition – one arm of a between-subject experiment (e.g. control vs. treatment).

Subagent – an independent Claude Code conversation spawned to do a self-contained piece of work (here: simulate one batch of subjects) and return a result.

Blinding – withholding which condition is “treatment” vs. “control” from the analyst until after data collection, to prevent bias in analysis or interpretation.

Balance check – a statistical test confirming that random assignment produced similar-looking groups across conditions (age, gender, etc.), which is a precondition for trusting a simple between-condition comparison.

Skill – a markdown file (SKILL.md) that packages instructions for a recurring task, which Claude Code follows when the task matches.

Allocation – how a study’s total N is split across conditions (e.g. “equal” for an even split, or exact per-condition counts).

Positive-label coding – the explicit choice of which category of a binary outcome is coded 1 (vs. 0), made explicit rather than left to an alphabetical default so the sign of every effect is interpretable.

Balance imbalance (false positive) – a statistically significant difference between conditions on a variable that should, in truth, be balanced by randomization; expected to occur by chance some fraction of the time across many tests, so it should prompt a robustness check rather than an assumption of a broken sampler.

Effect size – the magnitude of a treatment’s impact (e.g. a regression coefficient), as distinct from its statistical significance (p-value); a result can be significant with a small effect size, or large but not significant at a small N.

Mediator / moderator (out of scope here) – a mediator is a variable that explains *why* an effect occurs (e.g. affordability perception explaining a framing effect); a moderator is a variable that changes *how strong* an effect is for different subgroups (e.g. financial literacy). Both require dedicated analyses (Query Theory / GSEM mediation, floodlight moderation) not included in this workflow’s v1 scope.

Replication assessment – classifying a result against a reference study as *replicates* (same sign, both significant), *partially replicates* (same sign, only one significant), or *does not replicate* (opposite sign, or a near-zero estimate where the reference was significant).

Command cheat sheet

Windows (PowerShell):

```
# Refresh PATH in a new PowerShell session
$machinePath = [System.Environment]::GetEnvironmentVariable("Path","Machine")
$userPath = [System.Environment]::GetEnvironmentVariable("Path","User")
$env:Path = $machinePath + ";" + $userPath
```

```
# Create and populate a venv
python -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r scripts\requirements.txt
```

```
# Git + GitHub
git init
git add .
git commit -m "message"
gh repo create <name> --private --source=. --remote=origin --push
git push
```

```
# pandoc conversions
pandoc doc.md -o doc.docx
pandoc doc.md -o doc.pptx
pandoc doc.md -o doc.pdf --pdf-engine=typst --resource-path=".;assets"
```

macOS (Terminal):

```
# Pick up Homebrew's PATH in a new shell (if freshly installed)
source ~/.zshrc
```

```
# Create and populate a venv
python3 -m venv .venv
.venv/bin/python -m pip install -r scripts/requirements.txt
```

```
# Git + GitHub
git init
git add .
git commit -m "message"
gh repo create <name> --private --source=. --remote=origin --push
git push
```

```
# pandoc conversions
pandoc doc.md -o doc.docx
pandoc doc.md -o doc.pptx
pandoc doc.md -o doc.pdf --pdf-engine=typst --resource-path=".:assets"
```

7. Appendix: Bootstrapping Your Own N-Stage Workflow

Everything above assumes the skills in Section 3 already exist – because they do, in this project. If you're setting this up for a *different* research topic from a bare repository, this section shows

how those skills get created in the first place, using the same planning-first approach as the rest of this workbook.

7.1 Plan all 9 stages before building any of them

Describe your own version of the full pipeline to Claude Code – including stages you don’t intend to build yet. The goal at this point is a shared understanding of scope, not code.

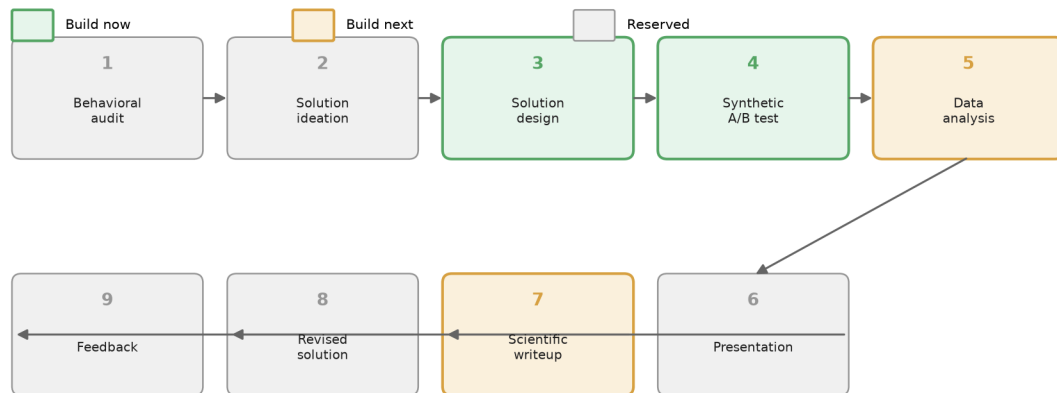
Copy-paste prompt template:

I’m setting up a 9-stage research workflow for [YOUR RESEARCH AREA], similar in spirit to: (1) audit, (2) ideation, (3) design, (4) synthetic test, (5) analysis, (6) presentation, (7) writeup, (8) revision, (9) feedback.

Help me sketch what each stage would need to do for my specific research area, before we build anything. For each stage: what are the inputs, what would a reasonable output look like, and how much of it seems automatable vs. requiring your judgment as the researcher each time?

What you get back: a stage-by-stage sketch, not code – read it and note which stages feel well-specified (clear inputs -> clear outputs, low ambiguity) versus which still feel fuzzy. That distinction drives the next step.

7.2 Make the scoping call



The roadmap this project actually used (Stages 3-4 built first, Stages 5 and 7 built next once validated, the rest reserved):

	Stage	Status	Why
3.	Design	Build now	Well-specified inputs (control/treatment stimulus, population, outcome); low ambiguity
4.	Synthetic test	Build now	Mechanical once Stage 3 exists: sample, assign, simulate
5.	Analysis	Build next	Depends on Stage 4's output existing and being trustworthy first
7.	Writeup	Build next	Depends on Stage 5's output; assembling, not deciding
1.	Audit	Reserved	Needs open-ended judgment about what's worth testing – hard to specify as a fixed procedure
2.	Ideation	Reserved	Same – creative/divergent, not a fixed input-to-output shape
6.	Presentation	Reserved	Audience- and format-dependent; more conversational than proceduralizable
8.	Revision	Reserved	Depends on real-world feedback this workflow doesn't yet collect
9.	Feedback	Reserved	Same – needs a live deployment loop this v1 doesn't have

The general rule: build the stages with the clearest inputs -> outputs first; defer the ones that need open-ended judgment until the core loop is validated.

7.3 Bootstrap the Skill files for your first two stages

Point Claude Code at one of this project's real skills as a style reference, and ask it to draft a new one for your topic.

Copy-paste prompt template:

I want to create a Skill for [YOUR STAGE NAME, e.g. “behavioral design intake”]. Here’s an existing example to match the style and structure: [PASTE THE CONTENTS OF `.claude/skills/behavioral-design/SKILL.md`].

Please draft a new `SKILL.md` following this same pattern, for: [DESCRIBE YOUR STAGE’S REQUIRED INPUTS AND EXPECTED OUTPUT].

What you get back: a draft `SKILL.md` – review it critically, adjust the required-inputs list to match your actual research area, then save it to `.claude/skills/<your-skill-name>/SKILL.md`. Repeat for your second stage.

7.4 Validate before you scale

Before trusting a new skill on a real project, run it on a tiny throwaway example first:

- For a design-intake-style skill: give it a made-up, low-stakes stimulus and confirm the structured output looks right
- For a synthetic-test-style skill: run it at $N=5-10$ (see 4.1’s token table) and spot-check a few rows of output before scaling up
- Iterate the skill’s wording based on what comes back – this is the same pilot-before-scale discipline as Section 4’s critical-steps checklist, applied one level up, to the skill itself rather than the data collection

7.5 What happened next in this project

This project didn’t stop at Stages 3-4. Once they were validated end-to-end (the $N=20$ pilot, then the $N=400$ scale-up described in the Quick Start), the same bootstrap-then-validate pattern was repeated to add Stage 5 (analysis) and Stage 7 (writeup). Stages 1, 2, 6, 8, and 9 remain reserved – not abandoned, just not yet specified clearly enough to automate. The lesson holds at every level of this workflow: scope small, validate, then expand.

7.6 Using a Google stack instead, at no cost

Status check this before class. Updated **September 3, 2026**: this has now been piloted once, end-to-end, on a toy $N=10$ project (branch `antigravity-pilot` in this repo) – all four stages (design, synthetic test, analysis, writeup) completed, and every output was independently re-verified, including byte-for-byte identical results from re-running the actual analysis scripts by hand. That’s real signal, not just the conceptual mapping below. It is still a *single small-N run*: behavior at realistic scale ($N=100+$), many parallel batches, and free-tier quota limits under sustained use are all still unconfirmed – see `.agents/PILOT-NOTES.md` on that branch for the full dated log. This is also a product space that has already changed twice in 2026 (Google discontinued its previous free options – Gemini Code Assist for individuals, and Gemini CLI’s free “Login with Google” path – on June 18, 2026, replacing both with the product described below), so confirm current terms before relying on this for a class.

Everything in Sections 1-7.5 assumes Claude Code. If students don't have access to it, **Google Antigravity** is currently the closest free substitute – an agentic IDE (available as its own VS Code fork, or as an extension inside regular VS Code) with a free “Individual” plan that requires no paid subscription. Google doesn't publish exact numbers, describing the free tier only as “a meaningful quota, refreshed weekly” – plan for a lighter pilot (e.g. N=5-10, not N=400) if using it for Stage 4. What makes it a workable substitute is that its building blocks map closely onto this project's own, not just onto Claude Code's brand:

This project	Antigravity equivalent
VS Code + Claude Code	VS Code + the Antigravity extension, or Antigravity's own VS Code-fork IDE
<code>.claude/skills/<name>/SKILL.md</code> (name + description frontmatter, description drives auto-invocation, optional scripts/ folder)	<code>.agents/skills/<name>/SKILL.md</code> – same convention: description is required and drives semantic auto-invocation, scripts/ folder optional
Subagents (a batch of ~10 simulated subjects per call, each with isolated context)	Antigravity's dynamic subagents – the main agent spawns child agents on the fly, each with an isolated context window, running in parallel where possible
CLAUDE.md project-level standing instructions (not heavily used in this project)	GEMINI.md (global, at <code>~/.gemini/</code>) or a project-level <code>.agents/rules/</code> folder
Python, git/GitHub, pandoc	Unchanged – none of Sections 1-6 is model-specific

One real mechanical difference: Claude Code subagents are spawned inline from a single chat thread. Antigravity routes multi-agent work through a separate “Mission Control / Agent Manager” surface – dispatching Stage 4's parallel batches would look like assigning tasks in a small console, not typing one prompt into one conversation.

To switch: install the Antigravity extension in VS Code (or Antigravity's standalone IDE) and sign in with a personal Google account – no credit card needed for the free plan. Recreate your Stage 3/4 skill files at `.agents/skills/<name>/SKILL.md` instead of `.claude/skills/<name>/SKILL.md`, using the same bootstrap-prompt pattern from Section 7.3 (point the agent at an example skill, ask it to draft a new one in that style). Everything from Section 1.2 onward (Python, git, pandoc) stays exactly as written.

Or, skip rebuilding it: get this project's already-piloted version directly from Git. The `.agents/skills/` files described above already exist, and have already been piloted once, on a dedicated branch of this project's own repository:

```
git clone https://github.com/prof-stephenshu/behavioral-research-env.git
cd behavioral-research-env
git checkout antigravity-pilot
```

Then follow Sections 1.2-1.3 exactly as written (Python venv + requirements.txt) – environment setup doesn’t change based on which agent you use. Open the folder in VS Code with the Antigravity extension signed in, and ask it “*What skills do you have available for this project?*” before trying a real run – it should list all four by name. .agents/PILOT-NOTES.md on that branch has the full dated log of what’s been verified and what hasn’t.

Access note: this repository is currently **private** and access is managed via a collaborator list (chosen deliberately over making it public, at least for now).

Instructor side – adding a student: 1. You need the student’s **GitHub username** (not their email or name) – ask them for it, or have them create a free GitHub account first if they don’t have one. 2. Repo page -> **Settings** -> **Collaborators and teams** -> **Add people** -> enter their username -> confirm. (Equivalent via the gh CLI, if you prefer: `gh api --method PUT repos/prof-stephenshu/behavioral-research-env/collaborators/<their-username>`.) 3. GitHub emails them an invitation. It’s pending, not active, until they accept it – check **Settings** -> **Manage access** to see who’s accepted versus still pending.

Student side – accepting: 1. Look for an email from GitHub (“prof-stephenshu invited you to behavioral-research-env”) or a notification at github.com/notifications, and accept it. 2. Once accepted, `git clone https://github.com/prof-stephenshu/behavioral-research-env.git` will work with your own GitHub credentials (the clone URL doesn’t change – access is what’s gated, not the link). 3. If clone still fails with a permission/404-style error, the invitation likely hasn’t been accepted yet – check step 1 again before assuming something is broken.

If you’d rather skip the agentic IDE entirely: Google’s raw Gemini API (via Google AI Studio) has a separate, more stable free tier with published limits (roughly 250-1,000 requests/day depending on the model, Flash-class models only, no credit card) that can be called directly from a Python script – the same shape as this project’s own opt-in “API mode” (Section 3.2). It has no agent, no Skills, and no terminal/file access of its own, but its terms are simpler and better-documented than Antigravity’s.